



Department of CSE- Data Science

1st Mini Project Presentation

**Project Name: Adaptive Memory Lifecycle Management
for Long-Running Agentic AI Systems**

By: Group-26
Prince Kumar
Kartikey
Mohd. Uzair

Adaptive Memory Lifecycle Management for Long-Running Agentic AI Systems

Problem Statement

Existing Agentic AI systems cannot efficiently manage long-term memory, making them less accurate, slower, and harder to scale over time.

Key Challenges

- **Limited Context:** Cannot remember all past interactions.
- **Static Memory:** Stores everything without proper organization.
- **Poor Retrieval:** May fetch irrelevant memories.
- **Duplicate Data:** Similar memories are stored repeatedly.
- **No Prioritization:** Important and unimportant memories are treated the same.
- **No Intelligent Forgetting:** Outdated memories are never removed.
- **No Compression:** Repetitive information is not summarized.
- **Low Memory Quality:** No check for accuracy or usefulness.
- **Scalability Issues:** Larger memory increases cost and response time.
- **Limited Adaptability:** Memory does not improve from user feedback or experience.

Proposed Topic

Adaptive Memory Lifecycle Management for Long-Running Agentic AI Systems

Objective

Develop an **adaptive memory framework** that **stores, retrieves, prioritizes, compresses, and forgets** information to improve long-term AI performance.

Key Features

- **Smart Storage** – Save only useful memories.
- **Memory Validation** – Keep accurate and reliable information.
- **Priority Ranking** – Rank memories by importance and recency.
- **Context Retrieval** – Fetch only task-relevant memories.
- **Memory Compression** – Merge repetitive information.
- **Intelligent Forgetting** – Remove outdated memories.
- **Adaptive Learning** – Improve memory using feedback.

Expected Outcome

Build a **scalable, efficient, and reliable memory framework** that improves **accuracy, reasoning, personalization, and response speed** in long-running AI agents.

Why This Topic?

As Agentic AI systems become more autonomous, **efficient memory management** is essential for building reliable and scalable AI agents.

Why This Research Matters

- **Growing Adoption** – Long-running AI agents are becoming common.
- **Limited Context** – LLMs cannot remember everything.
- **Memory Bloat** – Continuous storage reduces efficiency.
- **Poor Retrieval** – Irrelevant memories affect responses.
- **No Intelligent Forgetting** – Outdated memories remain stored.
- **Scalability Issues** – More memory increases cost and latency.
- **Research Gap** – Adaptive memory lifecycle is still underexplored.
- **Real-World Impact** – Improves accuracy, reasoning, and personalization.

Literature Review / Existing Work

Paper	Key Contribution	Limitation
MemGPT (Packer et al., 2023)	Introduced external memory for LLMs.	No adaptive memory lifecycle.
Survey of LLM Memory (Zhang et al., 2025)	Reviewed existing memory techniques.	No implementation framework.
Agentic Memory (Yu et al., ACL 2026)	Combined short-term and long-term memory.	No complete lifecycle management.
Memory for Autonomous Agents (Du, 2026)	Analyzed memory architectures and evaluation.	No practical adaptive solution.
Anatomy of Agentic Memory (Jiang et al., 2026)	Proposed memory taxonomy.	Limited focus on adaptive memory.
From Storage to Experience (Luo et al., 2026)	Explained memory evolution.	Survey only, no working framework.

Research Gap:-

Existing research emphasizes memory storage and retrieval, whereas adaptive memory lifecycle management—including validation, prioritization, compression, reflection, archival, and intelligent forgetting—remains largely unexplored.

Proposed Methodology

Workflow

1. **Memory Acquisition** – Capture user interactions and task results.
2. **Memory Validation** – Verify accuracy and remove invalid data.
3. **Memory Classification** – Categorize into episodic, semantic, and procedural memory.
4. **Priority Ranking** – Rank memories by importance, recency, and usage.
5. **Context Retrieval** – Retrieve the most relevant memories for the current task.
6. **Memory Compression** – Merge and summarize repetitive memories.
7. **Adaptive Forgetting** – Archive or remove outdated and low-value memories.
8. **Continuous Learning** – Update memory using user feedback and task outcomes.
9. **Performance Evaluation** – Measure accuracy, latency, memory usage, and task success.

Proposed Methodology

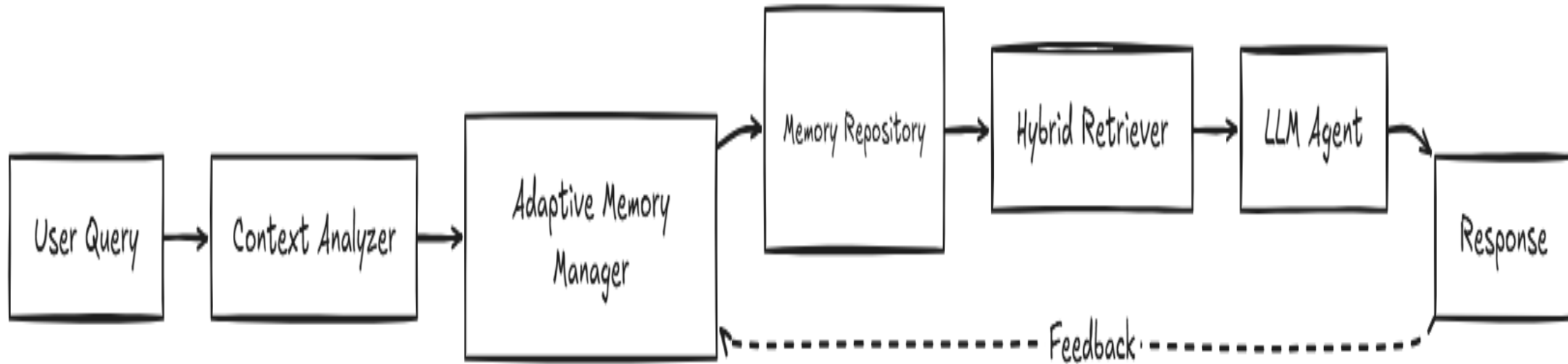


Fig: Architecture Diagram

Proposed Methodology

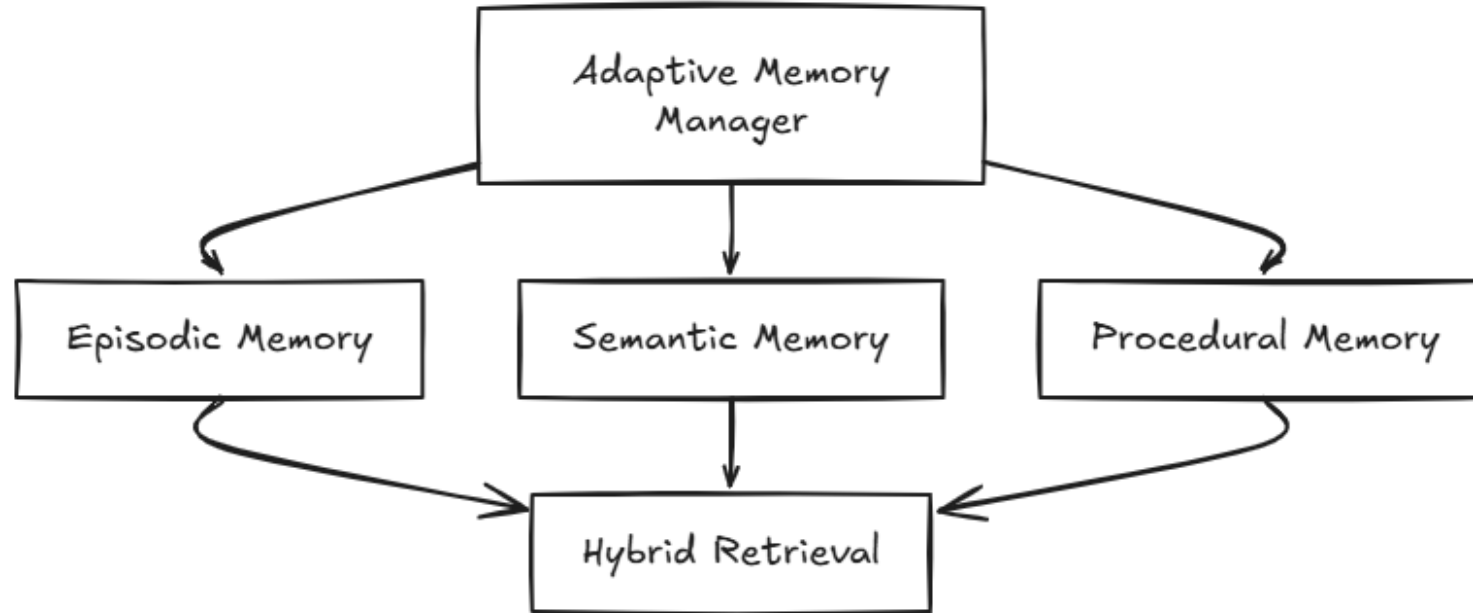


Fig: Memory Repository Detail

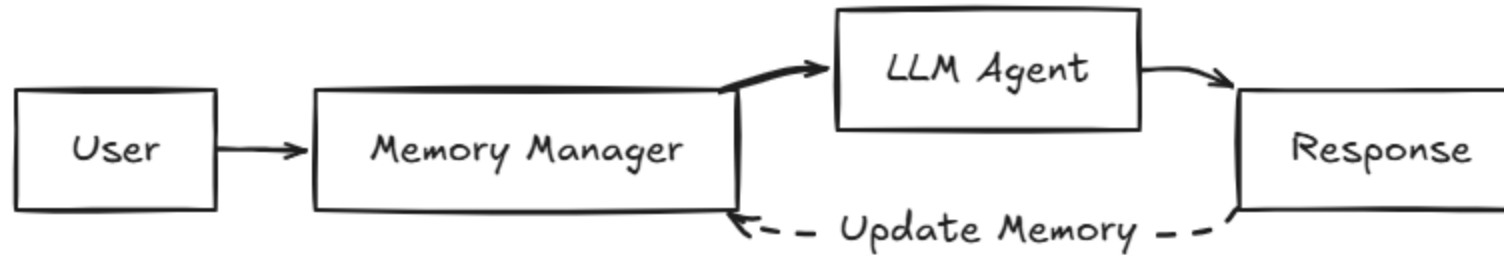


Fig: Overall Workflow

Expected outcomes

Develop an adaptive memory lifecycle framework that enables efficient, scalable, and reliable long-term memory management for Agentic AI systems.

Key Outcomes

- **Accurate Retrieval** – Fetch the most relevant memories.
- **Reduced Memory Bloat** – Remove duplicate and obsolete data.
- **Lower Cost** – Reduce token usage and storage requirements.
- **Faster Responses** – Improve retrieval speed and latency.
- **Better Long-Term Reasoning** – Retain useful knowledge over time.
- **Adaptive Learning** – Continuously improve using feedback.
- **Research Contribution** – A novel adaptive memory lifecycle framework for future AI systems.

References

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Thank You